Adarsha Balaji

Assistant Computer Scientist
Mathematics and Computer Science Division
Argonne National Laboratory
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Scientific Expertise

Neuromorphic computing, Spiking neural networks, High-performance computing, Reasoning Models, Multi-modal modals

Education

- Ph.D., Computer Engineering, Drexel University, Philadelphia, PA, USA 2017-2022
- Masters of Science, Computer Engineering, Drexel University, Philadelphia, PA, USA, 2015-2017
- Bachelor of Engineering, Telecommunication, Visvesvaraya Technological University, India 2008-2012.

Positions Held

Assistant Computer Scientist May 2025 - present
Mathematics and Computer Science Division,
Argonne National Laboratory, Lemont, Illinois
P.I. : Dr. Sandeep Madireddy

Post-doctoral Appointee May 2022 - May 2025
Mathematics and Computer Science Division,
Argonne National Laboratory, Lemont, Illinois
P.I. : Dr. Sandeep Madireddy and Dr. Prasanna Balaprakash

Neuromorphic RAMP Intern Summer 2021
Western Digital, Milpitas, CA

Doctoral Research Assistant Fall 2017 - Spring 2022
Thesis: Hardware-Software Co-Design for Neuromorphic Computing
Distributed, Intelligent and Scalable Computing Lab, Drexel University
Advisor : Dr. Anup Das

Senior Embedded Engineer June 2012 - June 2015
Robert Bosch Gmbh, India

Honors and Awards

- Drexel Dean's Fellowship, for outstanding academic performance from Drexel University, USA, from September 2015 to June 2017.

Research Products

Citations: 1051, h-index: 18, i10-index: 20 (Source: Google Scholar; 01/29/2026)

Refereed Journal Articles

1. **A. Balaji**, S. Song, A. Das, J. Krichmar, N. Dutt, J. Shackleford, N. Kandasamy, and F. Catthoor. Enabling Resource-Aware Compilation of Spiking Neural Networks to Neuromorphic Hardware IEEE Embedded Systems Letters (ESL), 2020, **Impact Factor: 1.52**
2. **A. Balaji**, S. Song, A. Das, N. Dutt, J. Krichmar, N. Kandasamy, and F. Catthoor. A Framework to Explore Workload-Specific Performance and Lifetime Trade-offs in Neuromorphic Computing. IEEE Computer Architecture Letters (CAL), 2019, **Impact Factor: 2.12**.
3. **A. Balaji**, A. Das, Y. Wu, K. Huynh, F. G. DellAnna, G. Indiveri, J. L. Krichmar, N. D. Dutt, S. Schaafsma, and F. Catthoor. Mapping spiking neural networks to neuromorphic hardware. IEEE Transactions on Very Large Scale Integration (TVLSI), 2019, **Impact Factor: 2.78**.
4. **A. Balaji**, F. Corradi, A. Das, S. Pande, S. Schaafsma, and F. Catthoor. Power-Accuracy Trade-Offs for Heartbeat Classification on Neural Networks Hardware. Journal of Low Power Electronics (JOLPE), 2018.
5. R. Kuttappa, **A. Balaji**, VsPano, Baris Taskin, and Hamid Mahmoodi. RotaSYN: Rotary Traveling Wave Oscillator SYNthesizer. IEEE Transactions on Circuits and Systems I: Regular Papers (TCAS-1), 2019, **Impact Factor: 4.14**

Publications in Refereed Proceedings of Conference and Workshop

6. **A. Balaji, Le Chen, Rajeev Thakur, Franck Cappello, and Sandeep Madireddy** Evaluation of Test-Time Compute Constraints on Safety and Skill Large Reasoning Models. Proceedings of the SC'25 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis, 2025
7. **A. Balaji et al.** Evaluating the Safety and Skill Reasoning of Large Reasoning Models Under Compute Constraints. Efficient Reasoning Workshop, NeurIPS, 2025
8. **A. Balaji** and P. Balaprakash Application - Hardware Co-Optimization of Crossbar-Based Neuromorphic Systems. International Conference on Machine Learning and Applications (ICMLA), 2023
9. **A. Balaji** and P. Balaprakash Application - Hardware Co-Optimization of Crossbar-Based Neuromorphic Systems. International Conference on Machine Learning and Applications (ICMLA), 2023
10. **Balaji, A.**, Hadidi, R., Kollmer, G., Fouda, M. E., Balaprakash, P. Network architecture search of X-ray based scientific applications. arxiv, 2024
11. M. L. Varshika, **A. Balaji**, F. Corradi, A. Das, J. Stuijt and F. Catthoor, Design of Many-Core Big Little μBrains for Energy-Efficient Embedded Neuromorphic Computing. Design, Automation and Test in Europe Conference and Exhibition (DATE), 2022. **Acceptance rate: 22%**
12. S. Song, H. Chong, **A. Balaji**, A. Das, J. Shackleford, and N. Kandasamy. DFSynthesizer: Dataflow-based Synthesis of Spiking Neural Networks to Neuromorphic Hardware. ACM Transactions on Embedded Computing Systems, 2022.

13. S. Song, **A. Balaji**, A. Das, and N. Kandasamy, Design-Technology Co-Optimization for NVM-based Neuromorphic Processing Elements. ACM Transactions on Embedded Computing Systems (TECS), 2022.
14. **A. Balaji**, S. Song, T. Titirsha, A. Das, J. Krichmar, N. Dutt, J. Shackleford, N. Kandasamy, and F. Catthoor. NeuroXplorer 1.0: An Extensible Framework for Architectural Exploration with Spiking Neural Networks. International Conference on Neuromorphic Systems (ICONS), 2021.
15. T. Titirsha, S. Song, **A. Balaji**, and A. Das. On the role of system software in energy management of neuromorphic computing. In Proceedings of the 18th ACM International Conference on Computing Frontiers (CF '21). Association for Computing Machinery, 2021.
16. **A. Balaji** and A. Das, Compiling Spiking Neural Networks to Mitigate Neuromorphic Hardware Constraints. 11th International Green and Sustainable Computing Workshops (IGSC), 2020.
17. **A. Balaji**, Y. Wu, A. Das, F. Catthoor, and S. Schaafsma. Exploration of Segmented Bus As Scalable Global Interconnect for Neuromorphic Computing. Proceedings of the Great Lakes Symposium on VLSI (GLSVLSI), ACM, 2019.
18. **A. Balaji**, S. Ullah, A. Das, and A. Kumar. Design Methodology for Embedded Approximate Artificial Neural Networks. Proceedings of the Great Lakes Symposium on VLSI (GLSVLSI), ACM, 2019.
19. **A. Balaji** and A. Das. A Framework for the Analysis of Throughput-Constraints of SNNs on Neuromorphic Hardware. IEEE Computer Society Annual Symposium on VLSI (ISVLSI), 2019.

Thesis

20. **A. Balaji.** *Hardware-Software Co-Design for Neuromorphic Computing*. Ph.D. thesis, Drexel University, Philadelphia, Pennsylvania, USA 2022.

Talks and Posters Presented at Conferences and Workshops

1. **A. Balaji**. Energy Efficient Implementation of the Transformer Architecture using Spiking Neural Networks Laboratory for Applied Mathematics, Numerical Software, and Statistics (LANS), Argonne National Laboratory (ANL), 2024.
2. **A. Balaji**. Co-optimization of Spiking Neural Networks with Neuromorphic Hardware, IEEE Nanotechnology Materials and Devices Conference (NMDC), 2024.
3. **A. Balaji**. Co-optimization of Spiking Neural Networks with Neuromorphic Hardware, IEEE Workshop on Microelectronics and Electron Devices (WMED), 2024.
4. **A. Balaji**. Exploration of Segmented Bus As Scalable Global Interconnect for Neuromorphic Computing. Proceedings of the Great Lakes Symposium on VLSI (GLSVLSI), ACM, 2019.
5. **A. Balaji**. Design Methodology for Embedded Approximate Artificial Neural Networks. Proceedings of the Great Lakes Symposium on VLSI (GLSVLSI), ACM, 2019.
6. **A. Balaji**. Heartbeat Classification in Wearables Using Multi-layer Perceptron and Time-Frequency Joint Distribution of ECG. IEEE/ACM international conference on Connected Health: Applications, Systems and Engineering Technologies (CHASE), 2018.
7. **A. Balaji**. Reliable Neuromorphic Computing. IBM IEEE CAS/EDS – AI Compute Symposium, 2018.

Software Developed

1. A. Balaji, H. D. Phung, S. Song and A. Das. NeuroXplorer1.0: A fast and extensible framework that is based on a generalized template for modeling a neuromorphic architecture that can be infused with the specific details of a given hardware and/or technology, 2020-2022 NeuroXplorer can perform both low-level cycle-accurate architectural simulations and high-level analysis with data-flow abstractions. NeuroXplorer's optimization engine can incorporate hardware-oriented metrics such as energy, throughput, and latency, as well as SNN-oriented metrics such as inter-spike interval distortion and spike disorder.
   https://github.com/drexel-DISCO/NeuroXplorer
2. A. Balaji, S. Song and A. Das. SpiNeMap: A design methodology to map SNNs to crossbar-based neuromorphic hardware, minimizing spike latency and energy consumption, 2019–2020. SpiNeMap operates in two steps: SpiNeCluster and SpiNePlacer. SpiNeCluster is a heuristic-based clustering technique to partition an SNN into clusters of synapses.
   https://github.com/drexel-DISCO/SpiNeMap
3. A. Balaji, P. Adiraju, H. J. Kashyap and A. Das. NeuroXplorer1.0: PyNN-based common Python programming interface for hardware-software co-simulation of spiking neural network (SNN), 2018-2020 Interface of PyNN to CARLsim, a computationally-efficient, GPU-accelerated and biophysically-detailed SNN simulator. PyCARL facilitates joint development of machine learning models and code sharing between CARLsim and PyNN users.. Integrates cycle-accurate models of state-of-the-art neuromorphic hardware such as DynapSE in PyCARL, to accurately model hardware latencies and energy consumption.
   https://github.com/adarshabalaji/PyCARL

Academic Experience

Teaching Assistant

Fall 2016 - 2022

- Advanced Computer Architecture - Graduate level course.
- Digital Systems Design - Graduate level course.
- Digital Logic Design - Sophomore level course.
- Design with Embedded Processors - Sophomore level course.
- Fundamentals of programming - Python - Sophomore level course.

Collaboration, Management, and Leadership

Supervision: Past

1. Hanh Do Phung, VIP Student, Drexel University
2. Himanshu Umare, VIP Student, Drexel University
3. Raymond Yung, VIP Student, Drexel University

Research proposals contribution

1. RTML: Small: Design of System Software to Facilitate Real-Time Neuromorphic Computing. Role: Researcher Assistant; Funding Agency: NSF; Duration: 2019-2023

Services to research community

1. Topic Editor, Frontiers in Neuroinformatics - Advanced Neural Network Modeling in Neuroscience and Brain Research, 2025
2. Reviewer, IEEE Transactions on Very Large Scale Integration (VLSI) Systems, 2020,2022.
   Reviewed 8 journals in 2 years.
3. Reviewer, IEEE Transactions on Very Large Scale Integration (VLSI) Systems, 2020,2022.
   Reviewed 8 journals in 2 years.
4. Reviewer, Neural Networks - Official Journal of the International Neural Network Society, European Neural Network Society and Japanese Neural Network Society, Impact Factor: 9.657
   Reviewed 3 journals in 2 years.

Leadership Roles

1. Board Member, Graduate Student Association (GSA), Drexel University, PA, USA.
   Governing body and planning board primarily for graduate students enrolled at Drexel University.
2. Vertically Integrated Projects (VIP), Research Programme, Drexel University, PA, USA.
   Guided 10 undergraduate and graduate students for early research explorations.

Skills

Languages: C, C++, SystemC, Perl, Python, CUDA, Pthread, Matlab, Cadence Skill, TCL, RTL, VHDL, Verilog.
Tools: Synopsys - Design Compiler, IC Compiler, PrimeTime, Virtuoso, Encounter, RTL Compiler, Mentor - Calibre Extraction, Noxim.
ML/AI Tools: Tensorflow, Keras, PyTorch, NEST, Brian, PyNN, CARLSim, Nengo, vLLM